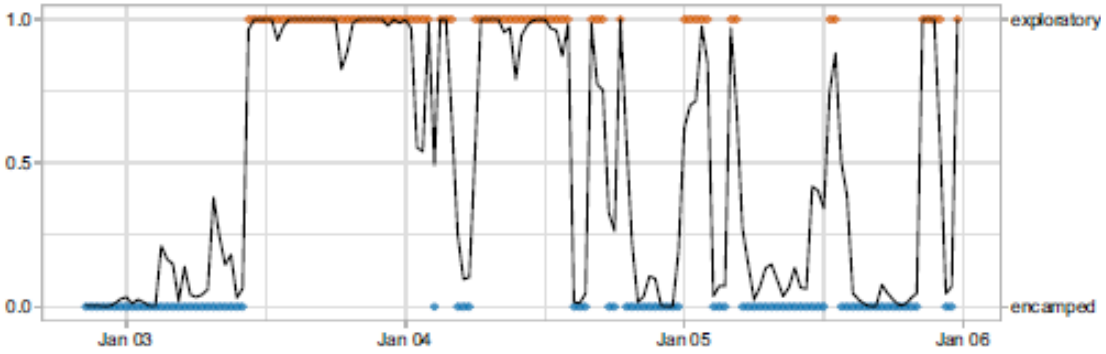
SMITHSONIAN'S NATIONAL ZOO & CONSERVATION BIOLOGY INSTITUTE

Introduction to Animal Movement Analyses

Categorizing Behavior States



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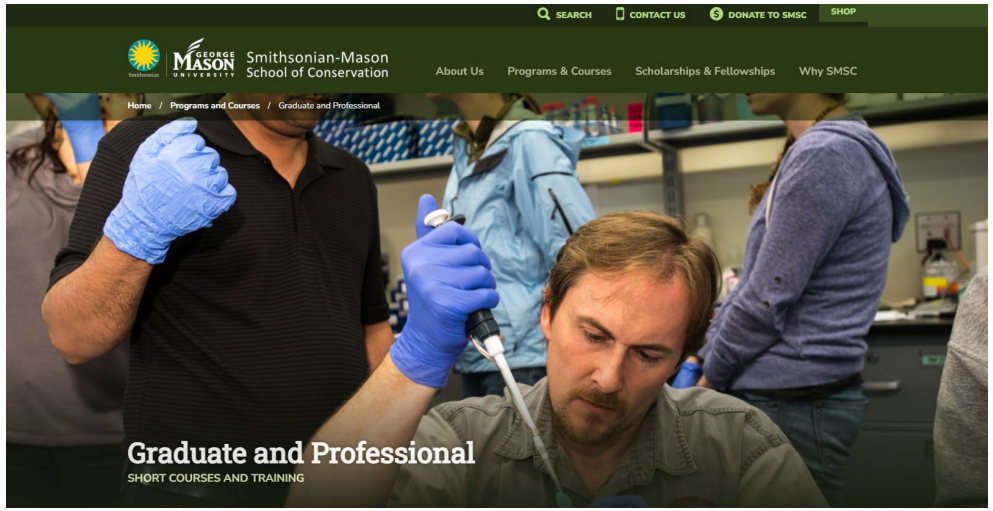
Conservation Biology Institute

$$\begin{array}{c|ccc} & S_1 & S_2 & S_3 \\ \hline S_1 & 0.8 & 0.1 & 0.1 \\ S_2 & 0.2 & 0.7 & 0.1 \\ S_3 & 0.05 & 0.3 & 0.65 \end{array}$$

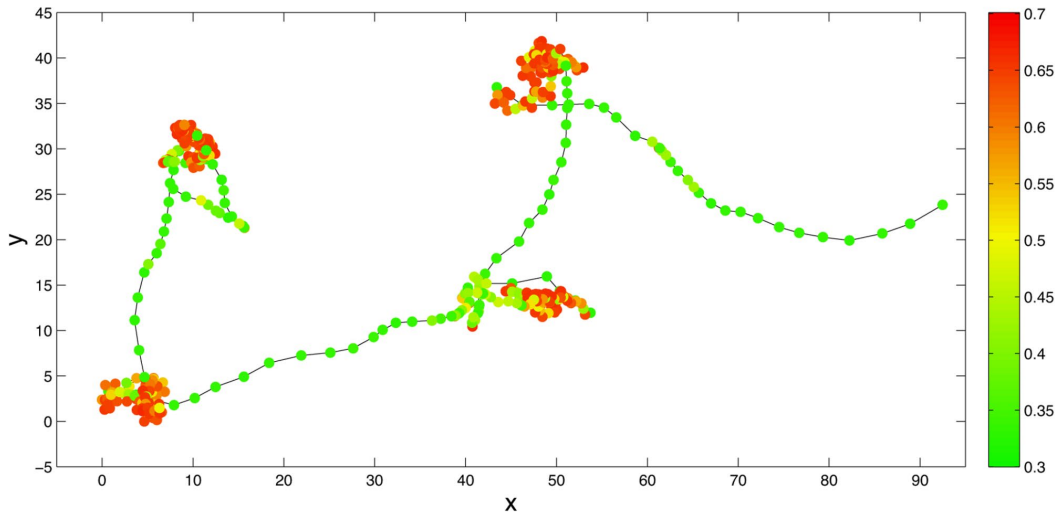
This lecture is based on a graduate training course taught at the Smithsonian-Mason School of Conservation
Many thanks to Dr. Joe Kolowski for sharing his notes.



https://smconservation.gmu.edu/programs/graduate-and-professional/

Behavioral Partitioning

- Hidden Markov Models (State-switching State Space Models)
- Behavioral Change Point Analysis (BCPA)
- First Passage Time/Residence Time
- Multi-state Random Walk



See Gurarie et al. 2016
for summary and comparison of methods

Applications of Behavior State Assignment

Exploratory
(What is the animal doing?)

- Metrics to describe and quantify the movement track
  - Are there distinct behaviors? If so, how many?
  - What distinguishes the behaviors?
  - When and where do the behaviors occur?

Explanatory
(Why is the animal doing it?)

- Associate behaviors and transitions with covariates
  - Season
  - Environmental cues
  - Predation or conspecifics

Predictive
(Can we anticipate movement from behavior?)

- If we modify the environment in some way, how will movement behavior change?

Hidden Markov Models

- Discrete time, discrete state, state space models
- Operate on **time-series data** for step lengths and turning angles
  - Could also be used on accelerometry data, but also occupancy data, capture-recapture data…
- Parameters of step length distribution and turning angle distribution determined by an underlying observed state

Hidden Markov Models

- Observation process and latent state process
  - Locations are the observation process, behavior state is the latent state process
  - Latent state process taken to be a Markov Chain
    - Discrete random variable, (1,2,…N)
    - N is # of potential states (typically <= 4)
  - Data distributions (step length and turn angle) vary by state
  - Transition probability matrix
    - Governs the sequence of switching most likely and how probable these switches are

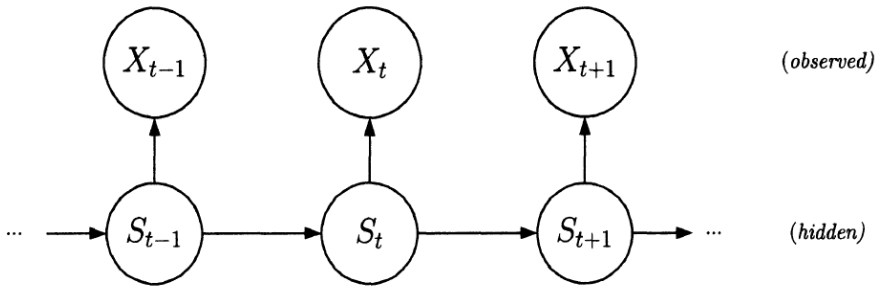


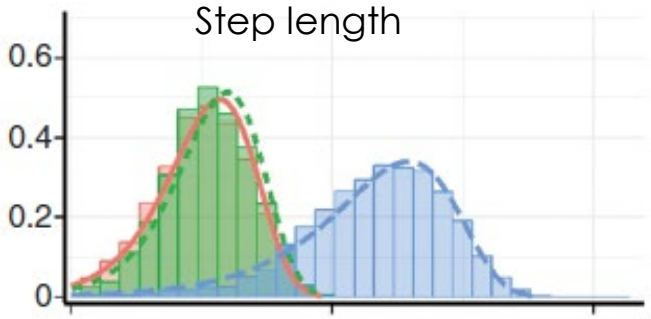
Figure 1. Dependence structure of an HMM in its most basic form.

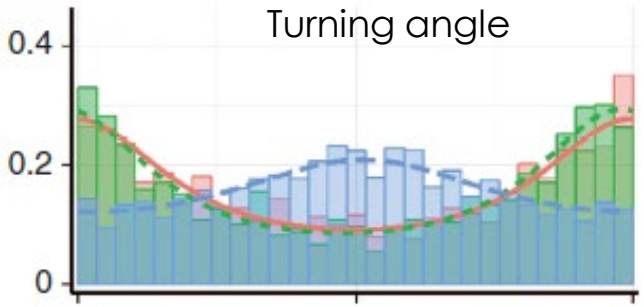
Pohle et al. 2017

Implementing HMMs in R

- Models estimate:
  1) Transition probabilities
  2) Mean and variance of step length and turn angle for all states
- Transition probabilities can be modeled as function of covariates

$$\begin{array}{c} \\ S_1 \\ S_2 \\ S_3 \end{array} \begin{array}{c} \begin{array}{ccc} S_1 & S_2 & S_3 \end{array} \\ \begin{pmatrix} 0.8 & 0.1 & 0.1 \\ 0.2 & 0.7 & 0.1 \\ 0.05 & 0.3 & 0.65 \end{pmatrix} \end{array}$$





HMM Parameters for Animal Movement

- N = # of states
- Starting state (typically taken to be equally likely)
- Distribution of step lengths (for each state)
  - Gamma, Weibull, log-normal and exponential distributions
  - Mean
  - Variance (SD)
  - Zero-mass
- Distribution of turning angles (for each state)
  - von Mises and wrapped Cauchy distributions
  - Mean
  - Clustering parameter
- Transition probability matrix

$$\begin{array}{c c} & \begin{array}{ccc} S_1 & S_2 & S_3 \end{array} \\ \begin{array}{c} S_1 \\ S_2 \\ S_3 \end{array} & \begin{pmatrix} 0.8 & 0.1 & 0.1 \\ 0.2 & 0.7 & 0.1 \\ 0.05 & 0.3 & 0.65 \end{pmatrix} \end{array}$$

Implementing HMMs in R

- Connecting location data to latent states (decoding algorithms)
  - Viterbi
    - Finds most probable sequence of states that generated the data
  - Forward-backward algorithm
    - Computes marginal probability of each state at time *t*
  - Forward-filtering backward sampling algorithm
    - Generates sequences that could have resulted in observed state
- Stationary probabilities available

- Assume regular sampling with negligible error
  - Paths can be smoothed prior to analysis if error on same scale as step lengths
  - No date/time information is included
- User specifies # of state (<= 4) and models with varying states can be compared with AIC

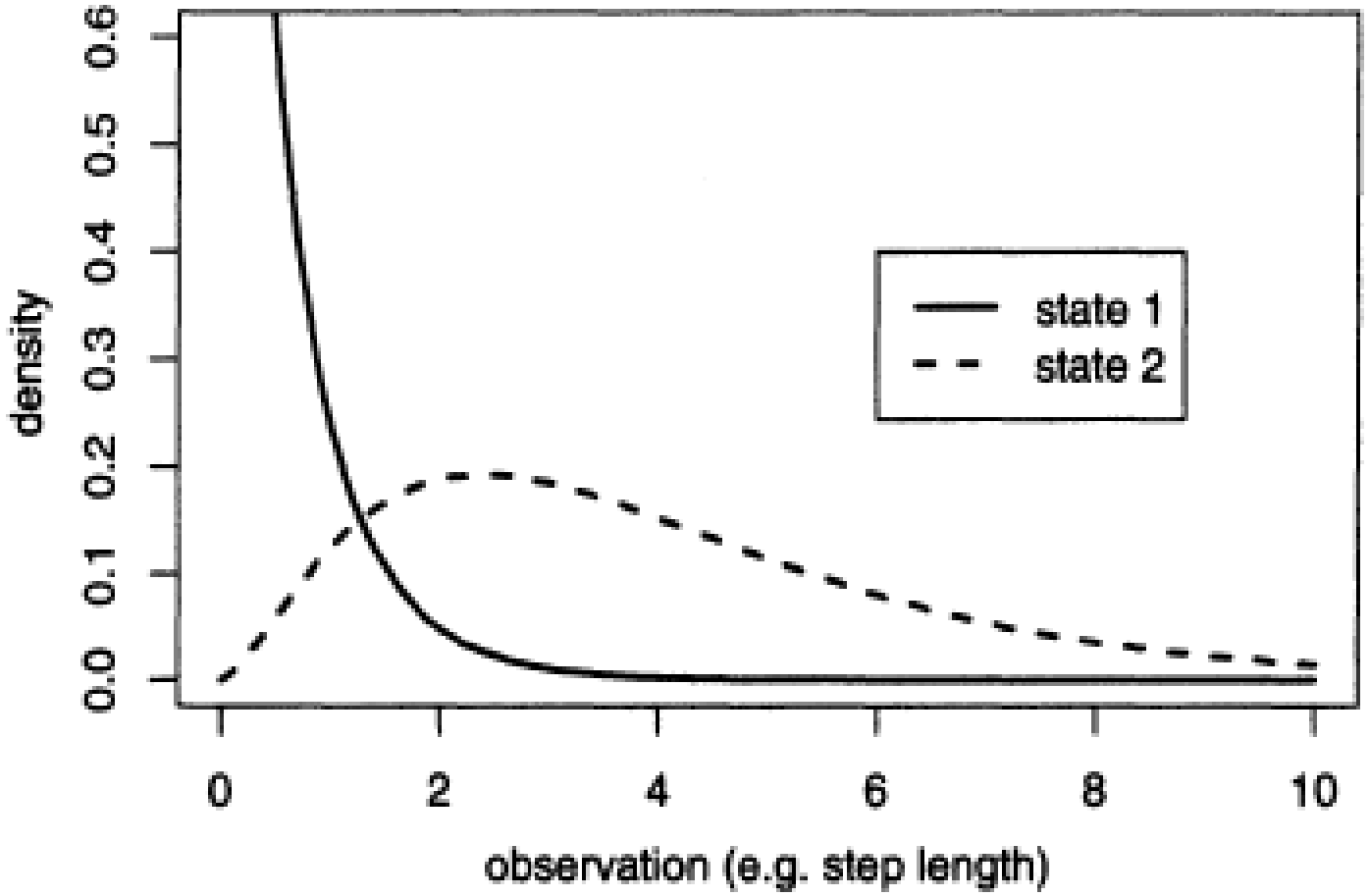


Figure 2. Gamma distributions of the baseline HMM used in the simulation experiments.

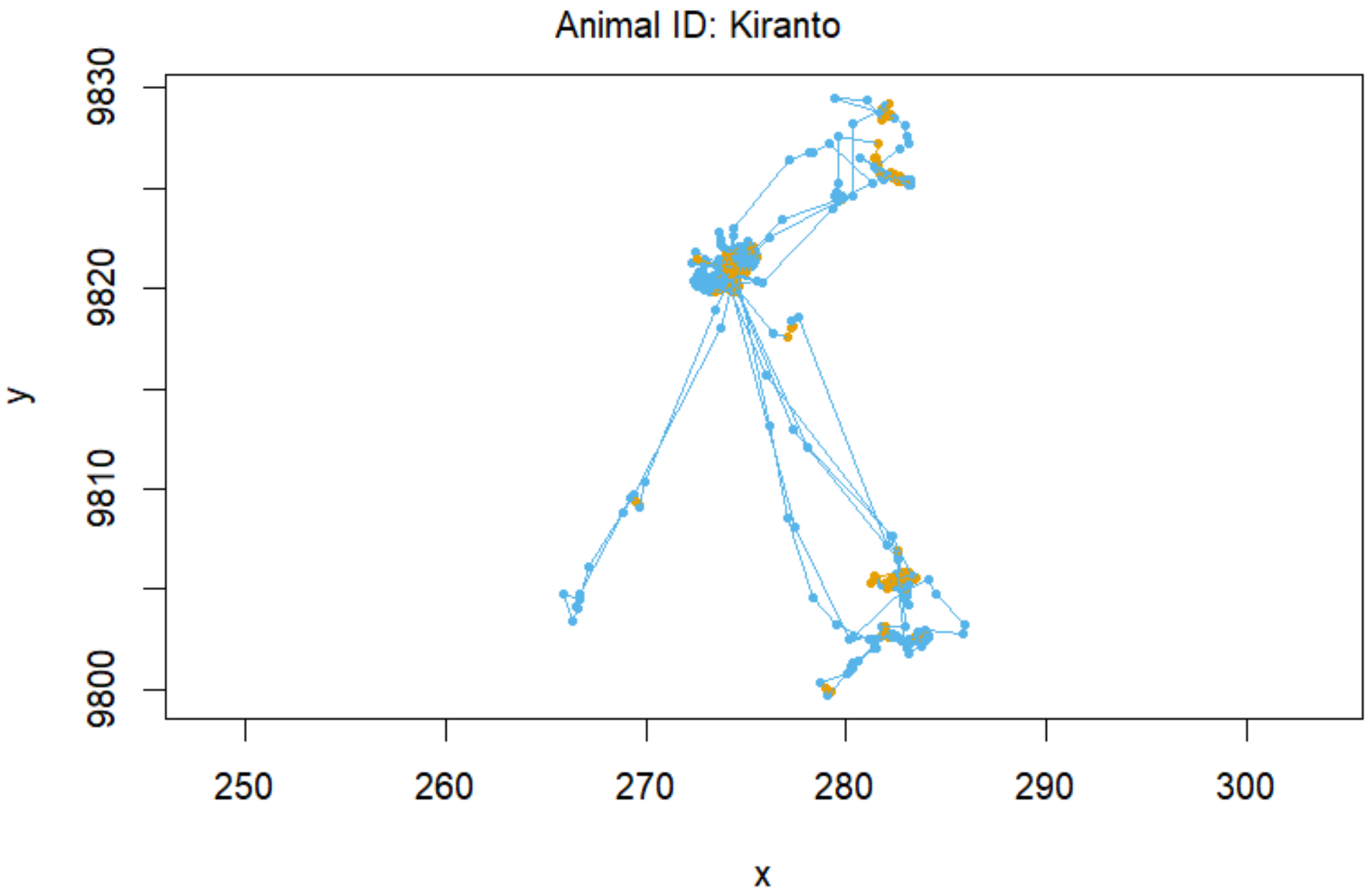
Pohle et al. 2017

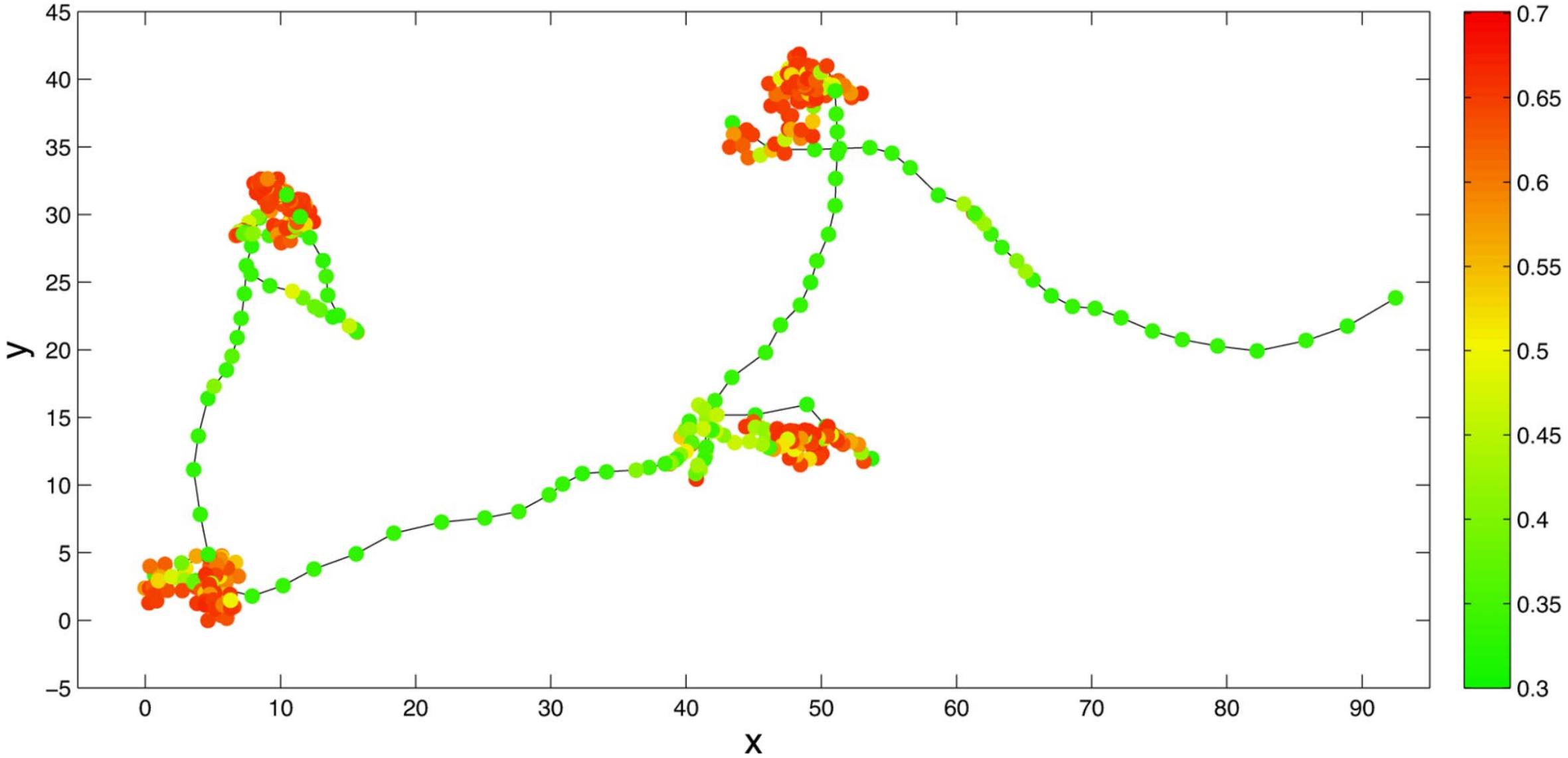
Hidden Markov Models with moveHMM

```
my_mod <- fitHMM(data = my_move_object,
                 nbStates= 2,                    # Needs to be specified (≤ 4)
                 stepPar0 = stepPar0,            # Starting values, step length distribution
                 anglePar0 = anglePar0,          # Starting values, turn angle distribution
                 formula = ~ 1)

plot(my_mod)
```

Ability to investigate impact of covariates on transition probabilities


Animal ID: Kiranto
9830
9820
9810
9800
y
250
260
270
280
290
300
x



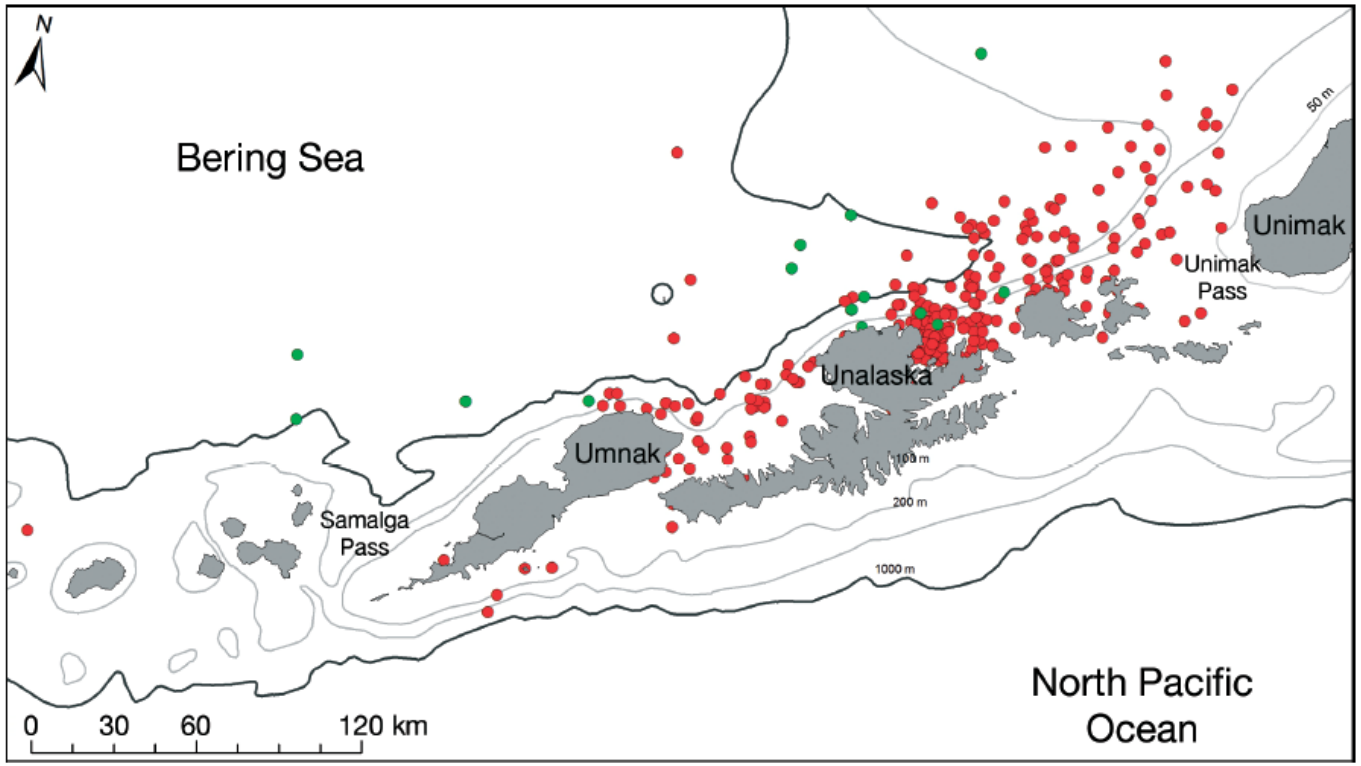
Postlethwaite and Dennis 2013

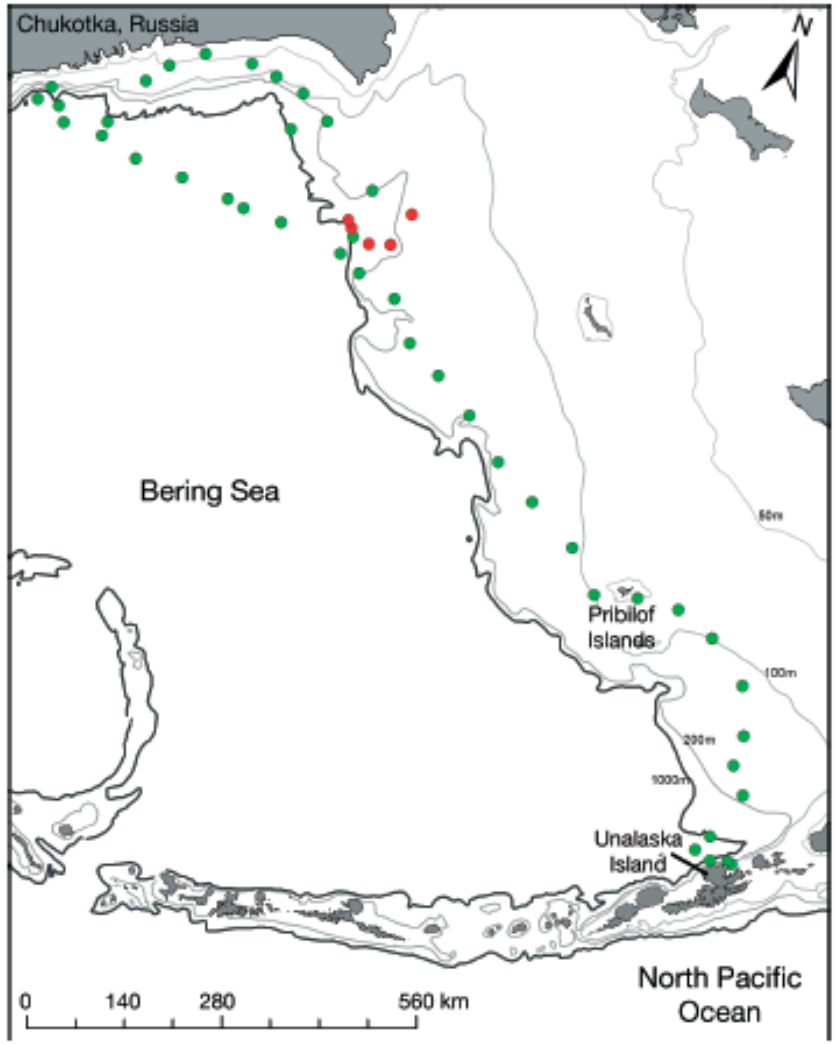
Challenges to HMMs

- Starting parameters values and correct distributional forms
- Choosing the # of states to model (Pohle et al. 2017)
- Impact of sampling schedule (Postlethwaite and Dennis, 2013)
  - Higher resolution tends to improve identification of behavioral states as long as error is much smaller than step length
- Irregular sampling
  - State space models with `crawl` or `aniMotum` to regularize

HMM Examples/Applications

Humpback whales





Individual variation in movements of satellite-tracked humpback whales *Megaptera novaeangliae* in the eastern Aleutian Islands and Bering Sea

Amy S. Kennedy[1,*], Alexandre N. Zerbini[1,2], Brenda K. Rone[1], Phillip J. Clapham[1]

HMM Examples/Applications

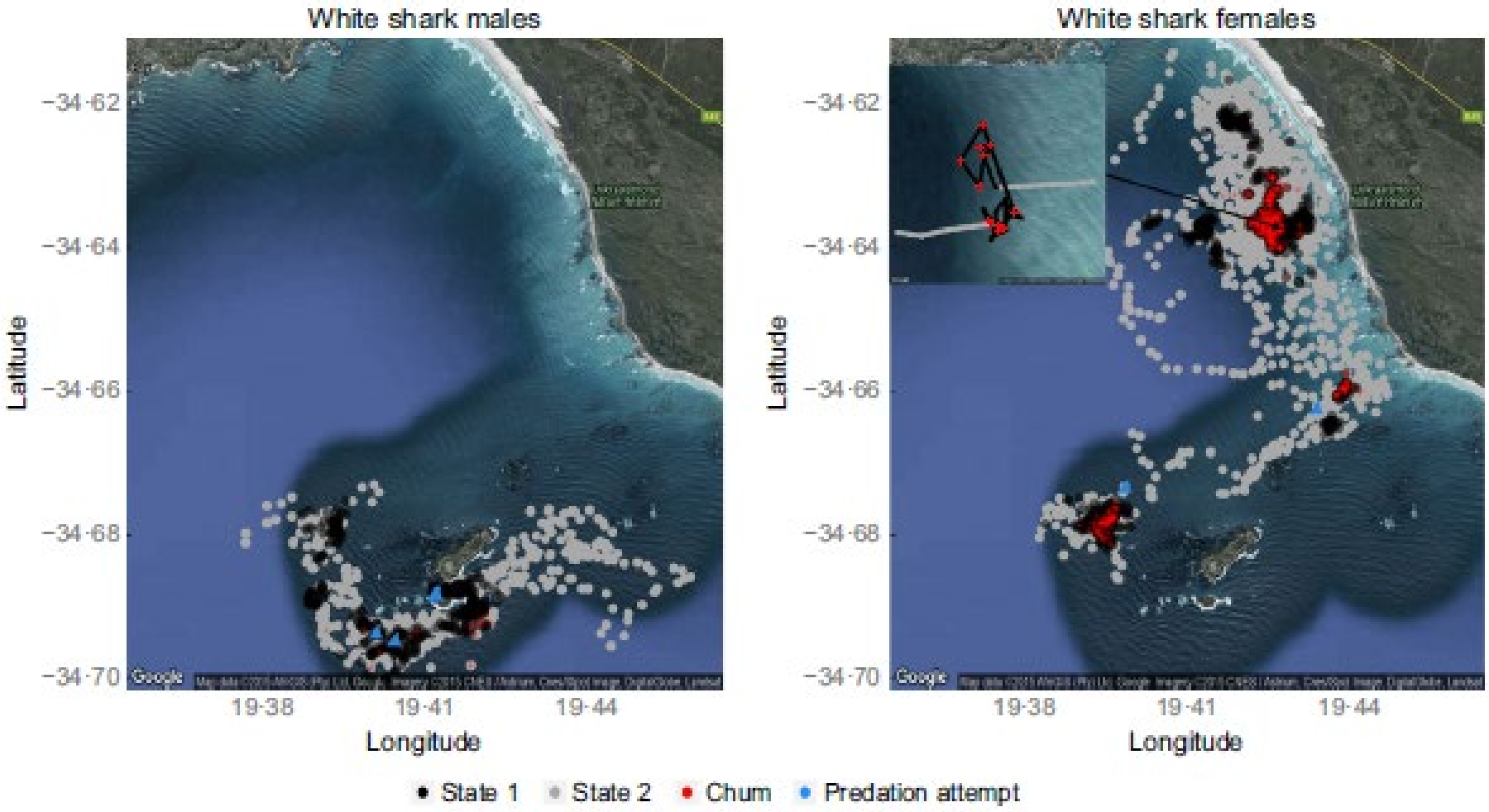
White sharks

Functional Ecology

Functional Ecology 2016, **30**, 1397–1407 doi: 10.1111/1365-2435.12613

Sex-specific and individual preferences for hunting strategies in white sharks

Alison V. Towner[*,1,2], Vianey Leos-Barajas[3], Roland Langrock[4,5], Robert S. Schick[4], Malcolm J. Smale[6,7], Tami Kaschke[2,8], Oliver J. D. Jewell[2,9,10] and Yannis P. Papastamatiou[11]



HMM Examples/Applications

Black bears

BioOne COMPLETE

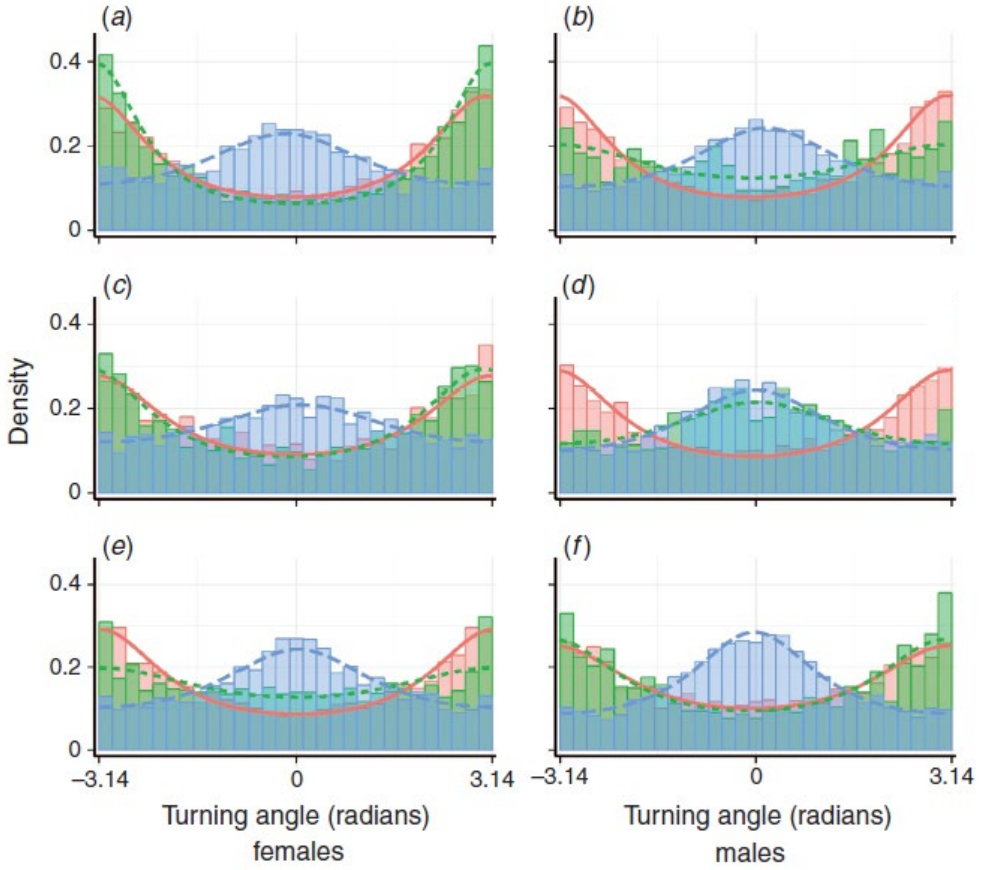
Incorporating movement patterns to discern habitat selection: black bears as a case study

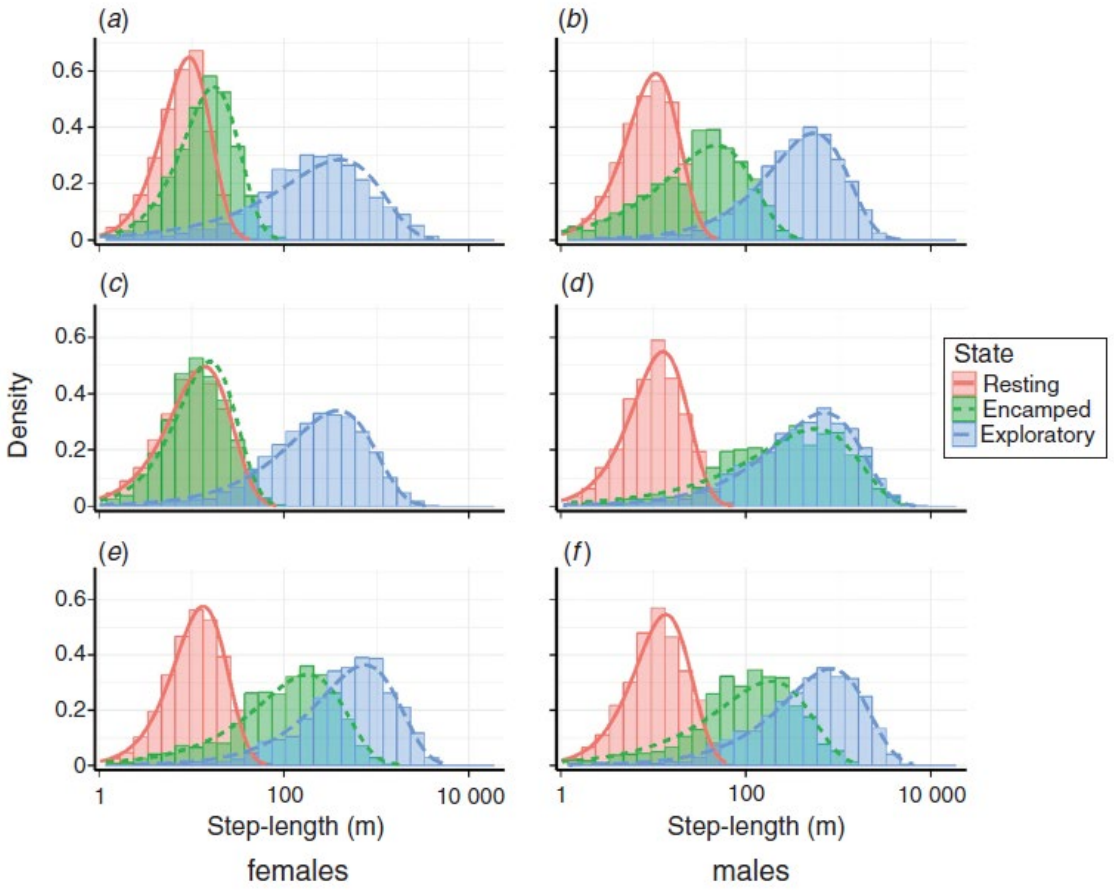
Authors: Karelus, Dana L., McCown, J. Walter, Scheick, Brian K., van de Kerk, Madelon, Bolker, Benjamin M., et al.

Winter

Summer

Fall





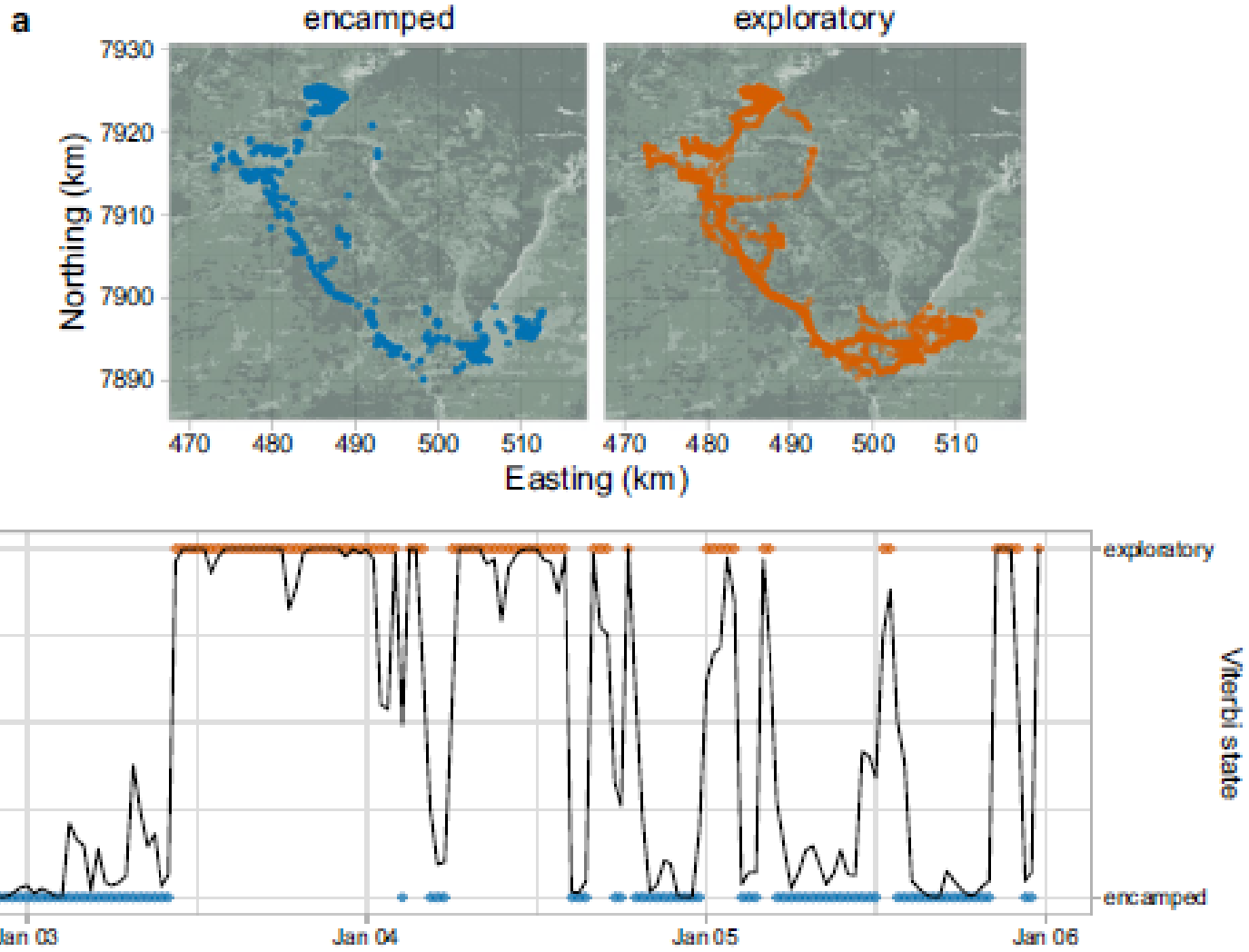
HMM Examples/Applications

METHODOLOGY Open Access

Flexible hidden Markov models for behaviour-dependent habitat selection

N. J. Klappstein[1,2*], L. Thomas[1] and T. Michelot[1,2]

Plains zebra



Behavior Change Point Analysis (BCPA)

- Identifies points in a time series when values undergo a significant shift
- Designed to identify changes in animal behavior – Gurarie et al. 2009
- Assumes continuous-space, continuous time stochastic process
- Uses maximum-likelihood to estimate parameters within a moving window
- Portions of path where values change abruptly identified as boundaries between movement modes

Summary of Analytical Approach (BCPA)

- Package `bcpa`
- Assume observations from continuous time process with mean, standard deviation and autocorrelation time scale
- Parameters vary according to an unknown number of states
- Select a response variable for analysis
  - Persistence velocity Vp=Vcos(θ) where V= speed and θ is turning angle (measures tendency to continue in same direction and speed)
  - Turning velocity Vt=Vsinθ (tendency to head in perpendicular direction)
  - Other variables could be used (e.g., V)

Summary of Analytical Approach (BCPA)

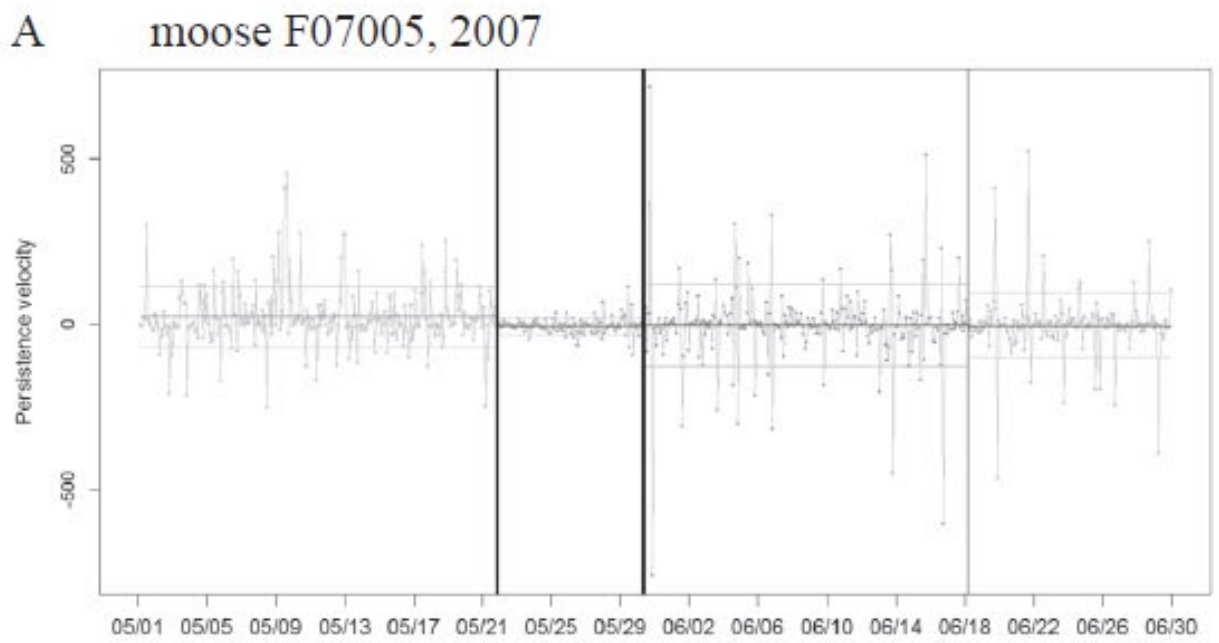
- Moving window of fixed size swept across data
- 3 parameters estimated
  - Mean, variance, autocorrelation time scale
- Identify locations in window which split data into two sets of the 3 parameters
  - Identify which of the three parameters describes the separation in the data
  - Could be null model, meaning no separation in any
- All change points recorded and parameters estimated on either side of change
- Which parameters change and how, which gives clue to the behavior change

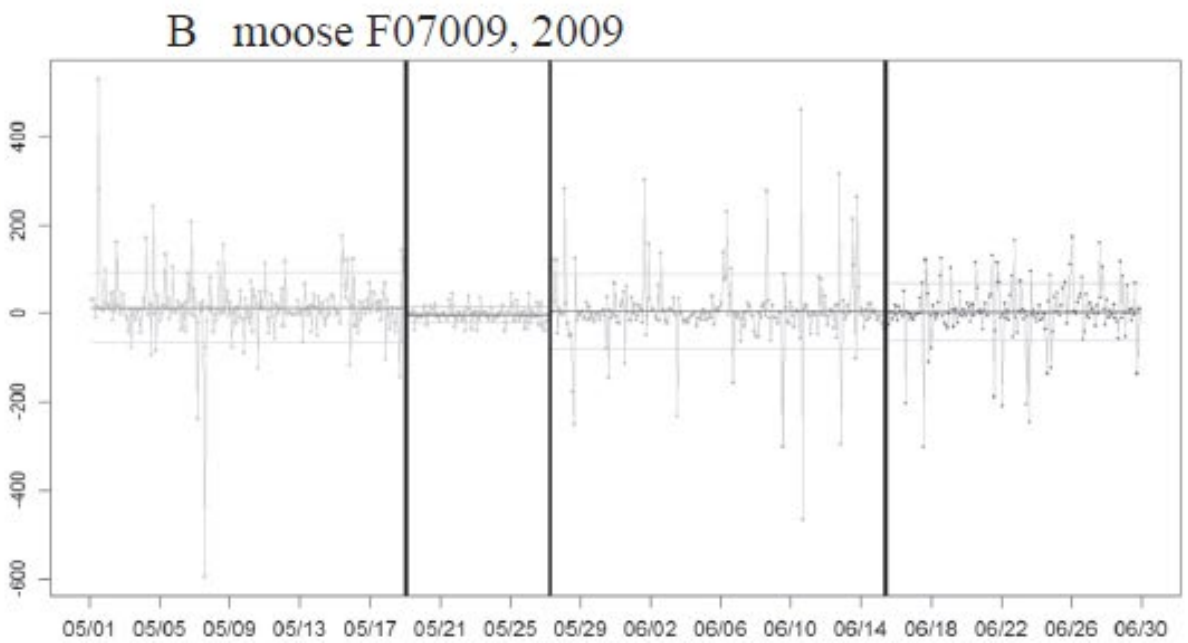
Key Points

- Larger window size more robust but more coarse
- Smaller window more sensitive but more likely to give spurious results
- Threshold parameter indicates how many windows must have selected the changepoint to be considered significant
  - Results in dropping less significant changepoints
- Flat vs. Smooth output
- Relatively complex model output and interpretation

BCPA Examples/Applications

Moose





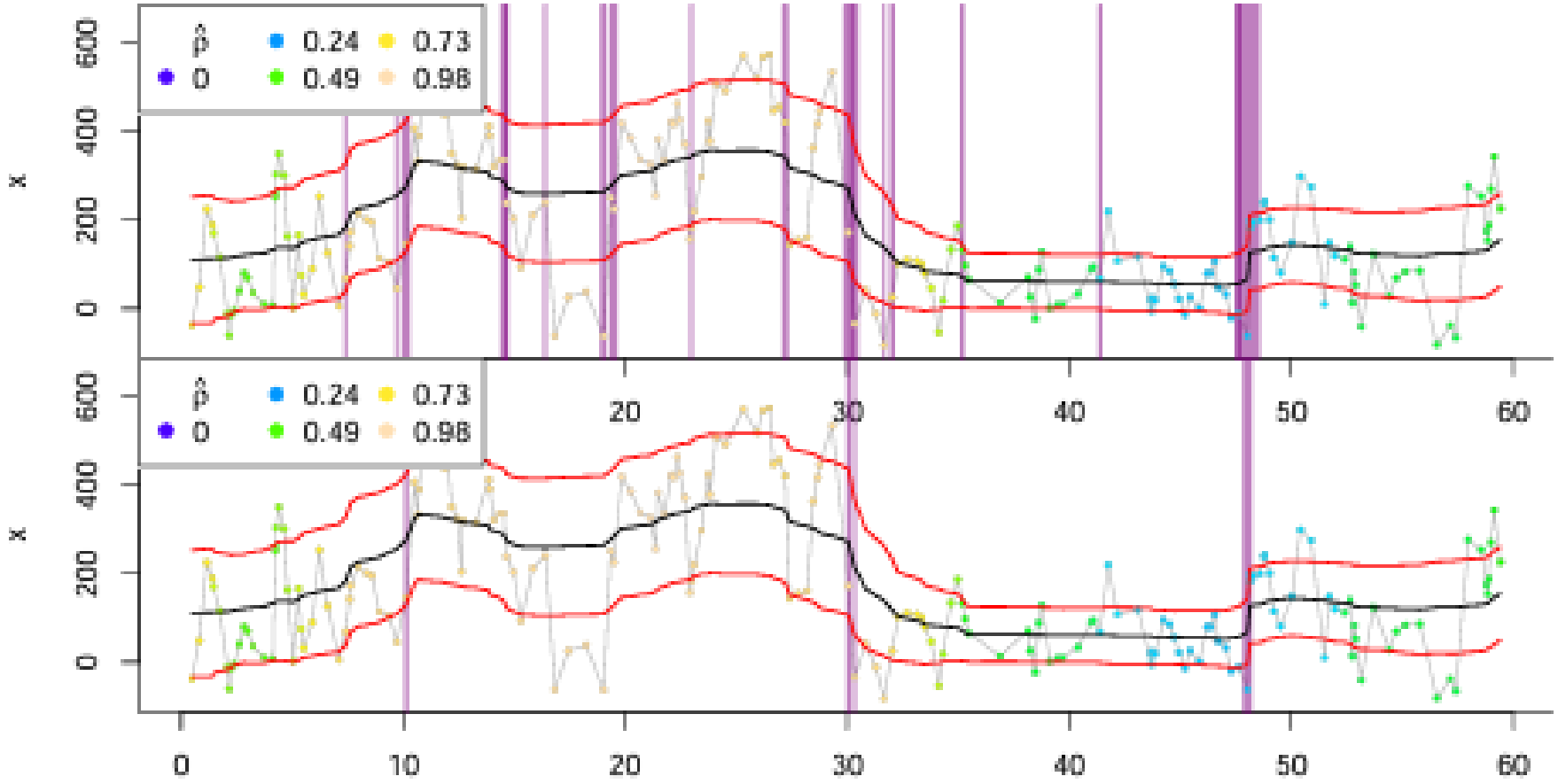
Original Articles

Using fine-scale movement patterns to infer ungulate parturition

Kerry L. Nicholson [a], Matthew J. Warren [b], Camille Rostan [c], Johan Månsson [c], Thomas F. Paragi [a], Håkan Sand [c]

BCPA Examples/Applications

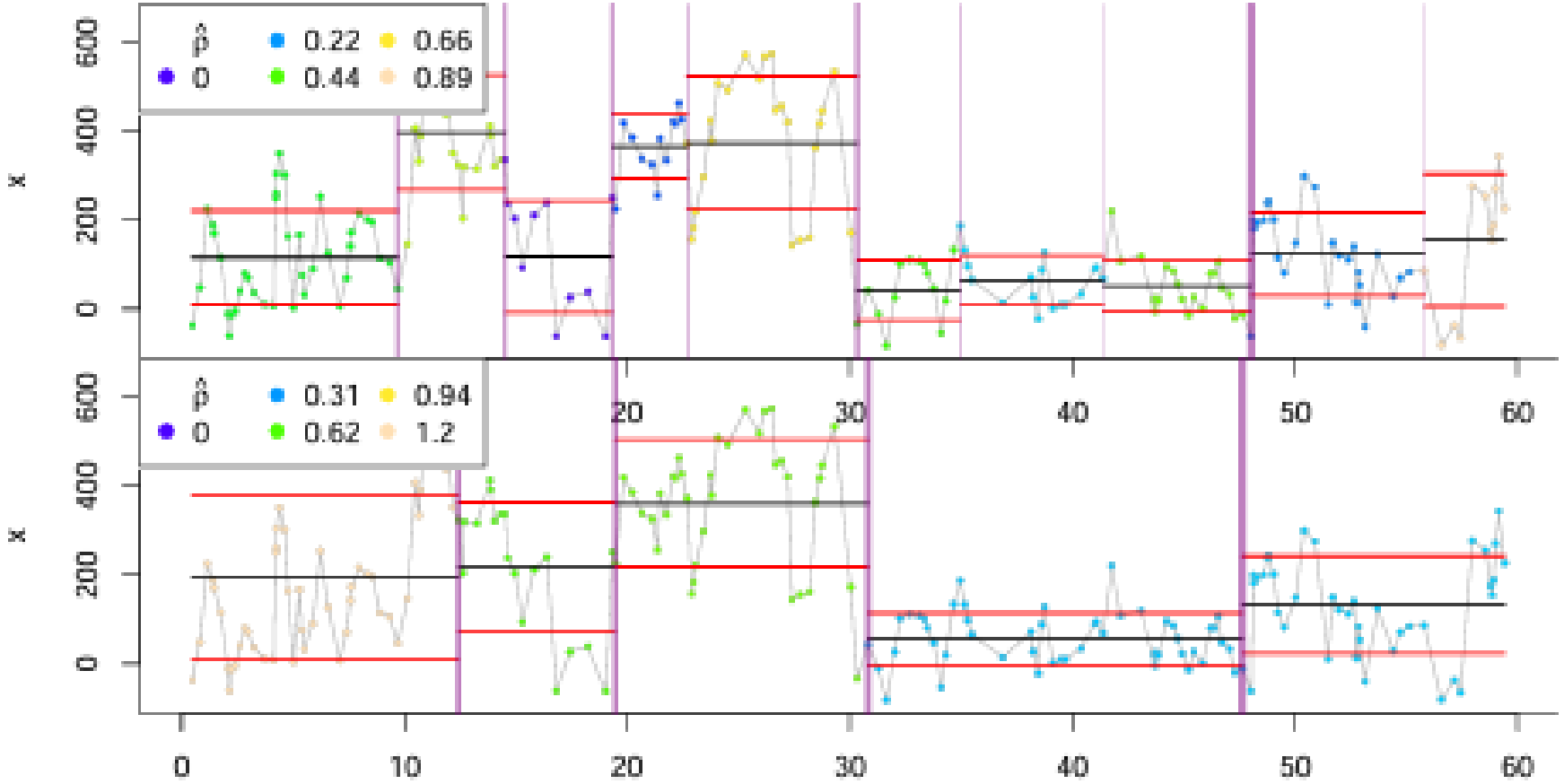
Flat vs Smooth output



See BCPA vignette for additional information

BCPA Examples/Applications

Flat vs Smooth output



See BCPA vignette for additional information

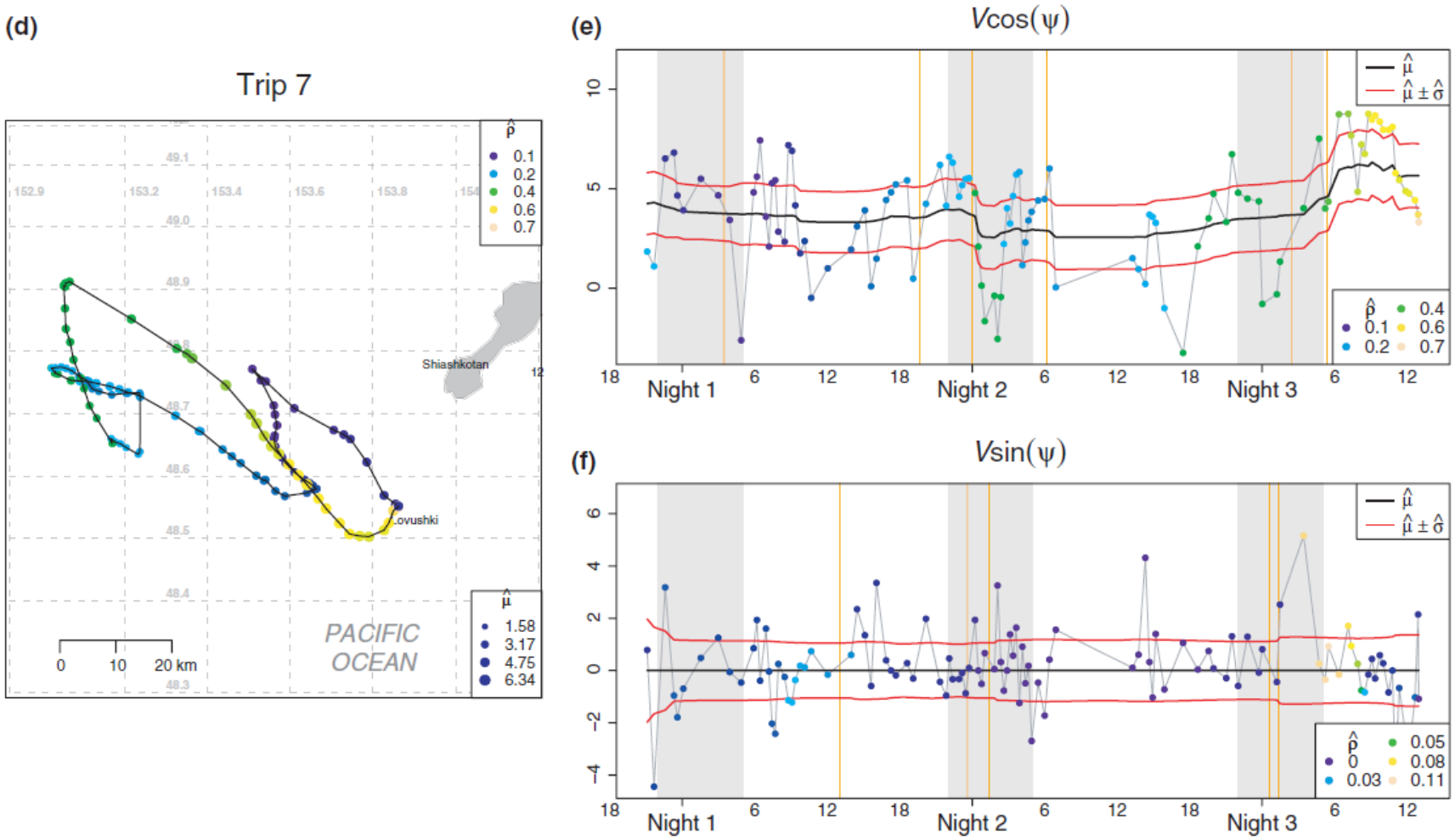
BCPA Examples/Applications

ECOLOGY LETTERS

A novel method for identifying behavioural changes in animal movement data

Eliezer Gurarie, Russel D. Andrews, Kristin L. Laidre

Fur seals



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